Java

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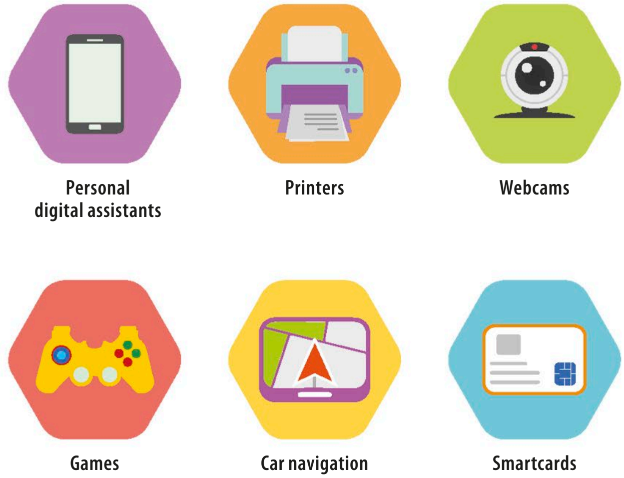
Java was developed in 1995 to make it easier to write code for the range of computers available at the time. It is still a major player today.

Background

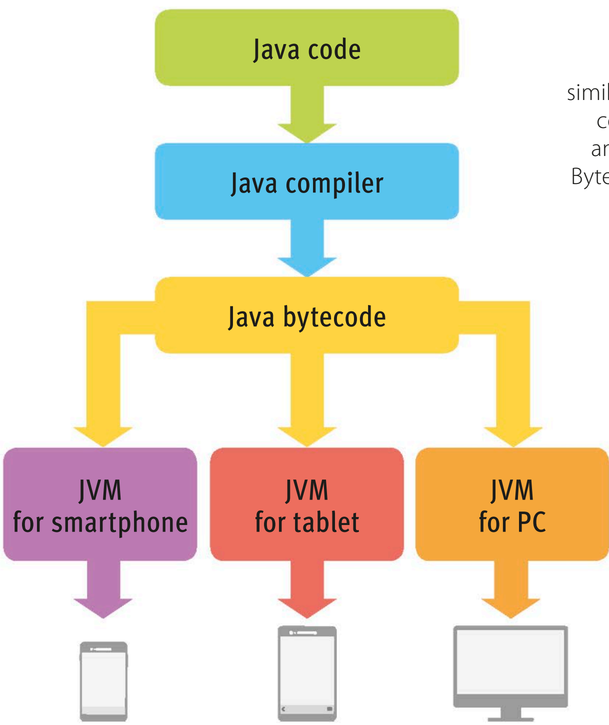
The Java programming language was developed by Canadian computer scientist James Gosling (b. 1955) for the American computer company Sun Microsystem's Java platform – a collection of software designed to allow programmers to develop a variety of systems. These ranged from tiny applications hosted on smartcards for personal banking, to large systems designed for use by many people across an organization. Web browsers soon included the ability to run small self-contained applications, called Java applets, which increased Java's popularity.



▷ **Language of gadgets**
The team behind Java wanted to design a language for programming the increasing number of electronic gadgets available, such as personal digital assistants (handheld personal computers) and webcams.

How does it work?

Java is an object-oriented language. Its syntax was designed to be similar to that of C and C++, but it doesn't include many of their low-level commands. A Java program is designed to behave in the same way on any machine. To enable this, a Java program is compiled into bytecode. Bytecode is machine code for the Java Virtual Machine (JVM), a simulated computer running on the user's real computer.



◁ **Java Virtual Machine**
The Java Virtual Machine is an abstraction that lets programmers write code without worrying about how it will work on a variety of different computers.

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Bytecode verification

Users can download files containing bytecode and run them on the JVM on their computer, which could allow malicious people to send out bytecode that could cause harm to computers. To avoid this, each bytecode file is examined by the JVM's bytecode verifier, which checks it doesn't perform specific undesirable actions, for example, accessing data to which it shouldn't have access to.